

Exercise 1: Implementing the Singleton Pattern

```
class Logger{
    private static Logger instance;
    private Logger(){}
    private static Logger getInstance(){
        if(instance==null){
            instance = new Logger();
        }
        return instance;
    }
    public void log(String message) {
        System.out.println("LOG: " + message);
    }
    public static void main(String[]args){
        Logger logger1 = Logger.getInstance();
        Logger logger2 = Logger.getInstance();

        logger1.log("First message");
        logger2.log("Second message");

        if (logger1 == logger2) {
            System.out.println("Same instance confirmed!");
        } else {
            System.out.println("Different instances");
        }
    }
}
```

Output:

FileEditSelectionViewGoRunTerminalHelp

CTS-JAVA-FSE-Exercises

Logger.java UFactoryMethodPattern.java U

Week1 > EngineeringConcepts > DesignPatterns > FactoryMethodPattern.java

```
DocumentFactory factory;

factory = new WordFactory();
Document doc1 = factory.createDocument();
doc1.open();

factory = new PdfFactory();
Document doc2 = factory.createDocument();
doc2.open();

factory = new ExcelFactory();
Document doc3 = factory.createDocument();
doc3.open();
}
```

PORTSTERMINALOUTPUTDEBUG CONSOLE

bash - DesignPatterns

Tharun@LAPTOP-1791NHQ27 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/DesignPatterns (main)
\$ javac Logger.java
Tharun@LAPTOP-1791NHQ27 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/DesignPatterns (main)
\$ java Logger
LOG: First message
LOG: Second message
Same instance confirmed!
Tharun@LAPTOP-1791NHQ27 MINGW64 /f/coding/CTS-JAVA-FSE-Exercises/Week1/EngineeringConcepts/DesignPatterns (main)
\$

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EXPLORER

CTS-JAVA-FSE-EXERCISES

Week1

EngineeringConcepts

DesignPatterns

Document.classU

DocumentFactory.classU

ExcelDocument.classU

ExcelFactory.classU

FactoryMethodPattern.classU

FactoryMethodPattern.javaU

Logger.classU

Logger.javaU

PdfDocument.classU

PdfFactory.classU

WordDocument.classU

WordFactory.classU